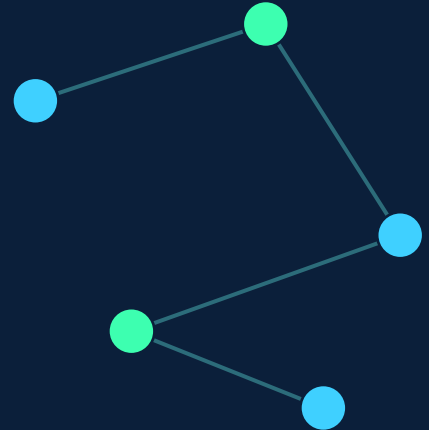


Opening a Secure Website: Tracing HTTPS Communication Using the TCP/IP Model

Junior Network Analyst Technical Presentation

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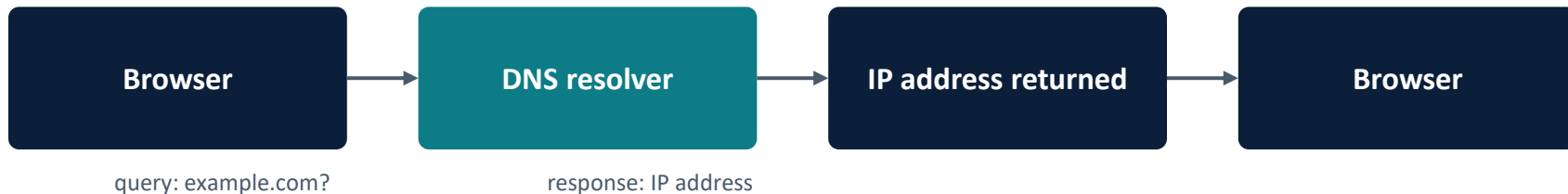




- The browser needs the destination IP address, not just the name.
- DNS resolution converts the domain name into an IP address.
- A transport connection is established to the server, usually on TCP port 443.
- TLS authenticates the server and protects the data in transit.
- The HTTPS request travels to the web server, and the response returns to the browser.

TCP/IP layer	OSI layers	Examples in this scenario
Application	7 Application, 6 Presentation, 5 Session	HTTPS (HTTP + TLS), DNS
Transport	4 Transport	TCP (port 443), UDP (port 53)
Internet	3 Network	IP, ICMP
Network Access	2 Data Link, 1 Physical	Ethernet, Wi-Fi, ARP

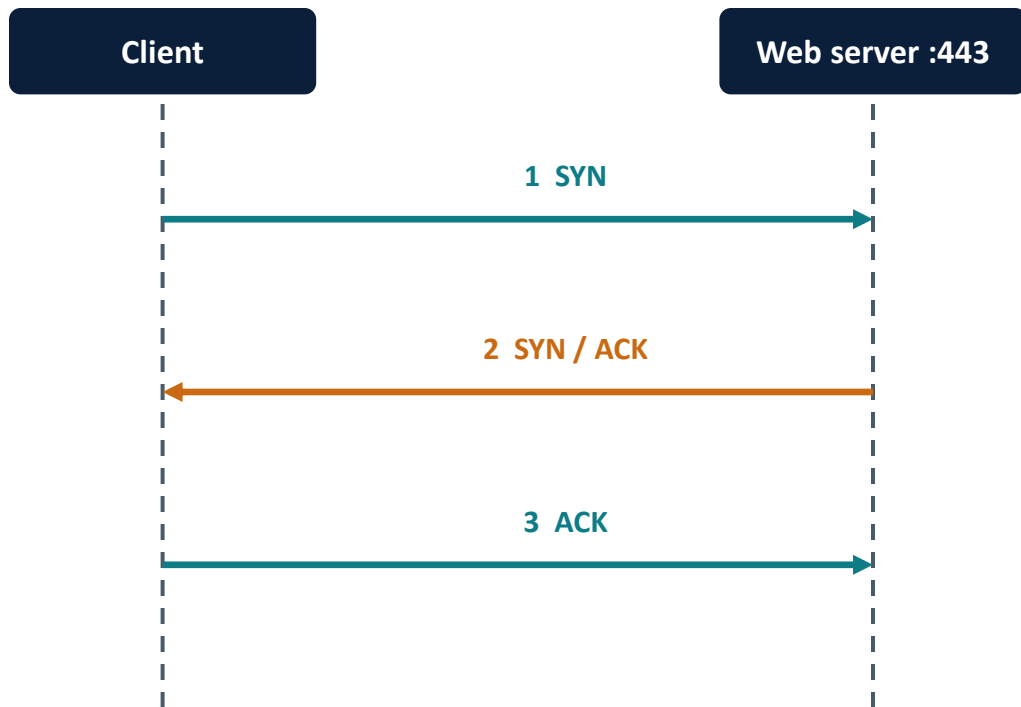
Why the difference? TCP/IP was built around working protocols, so functions OSI separates (encoding, session handling) are handled inside application protocols. OSI is a detailed reference model; TCP/IP is the practical model of the Internet.



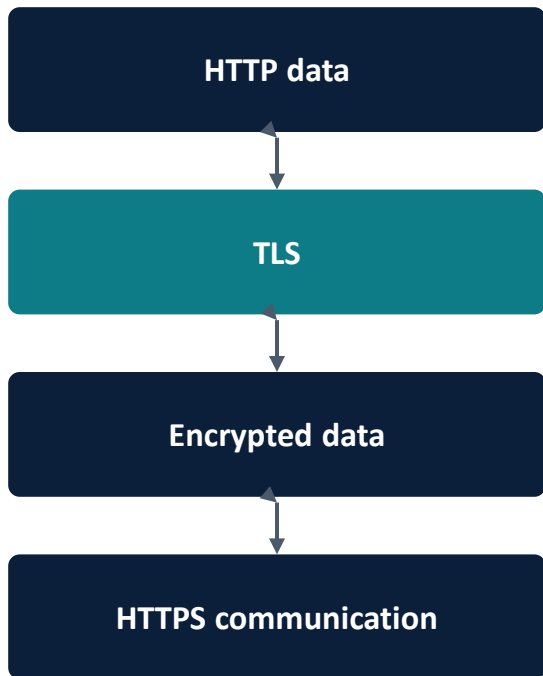
- DNS maps a domain name to an IP address; routing needs the IP.
- A DNS resolver answers the client query, asking other DNS servers if needed.
- DNS is an Application layer protocol.
- It commonly uses UDP port 53; TCP port 53 is also used in some cases (for example large responses).

TCP Connection Establishment

TRANSPORT LAYER



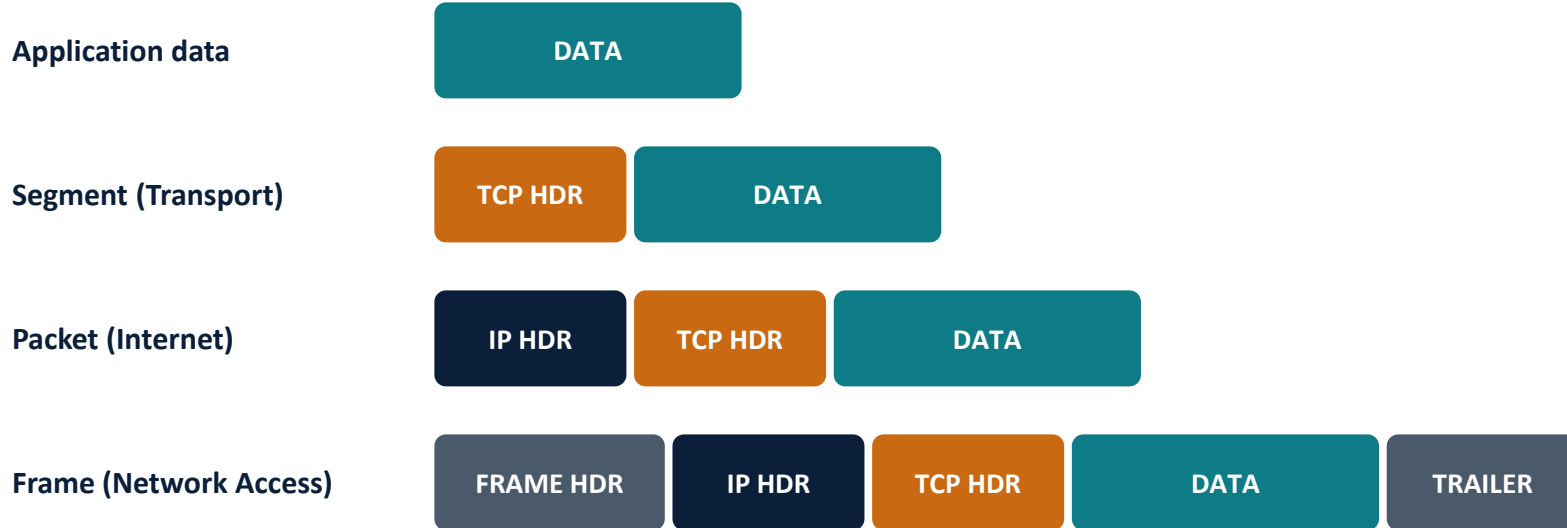
- SYN: client requests a connection.
- SYN/ACK: server accepts and acknowledges.
- ACK: client confirms.
- Connection established; data can flow.
- HTTPS over TCP uses destination port 443.



- HTTP defines web requests and responses (readable if not protected).
- TLS provides server authentication, encryption and integrity.
- HTTPS = HTTP carried over TLS.
- TCP does not encrypt; TLS provides the security.
- HTTP/1.1 and HTTP/2 commonly use TCP; HTTP/3 uses QUIC over UDP.

Encapsulation (Sender) and Decapsulation (Receiver)

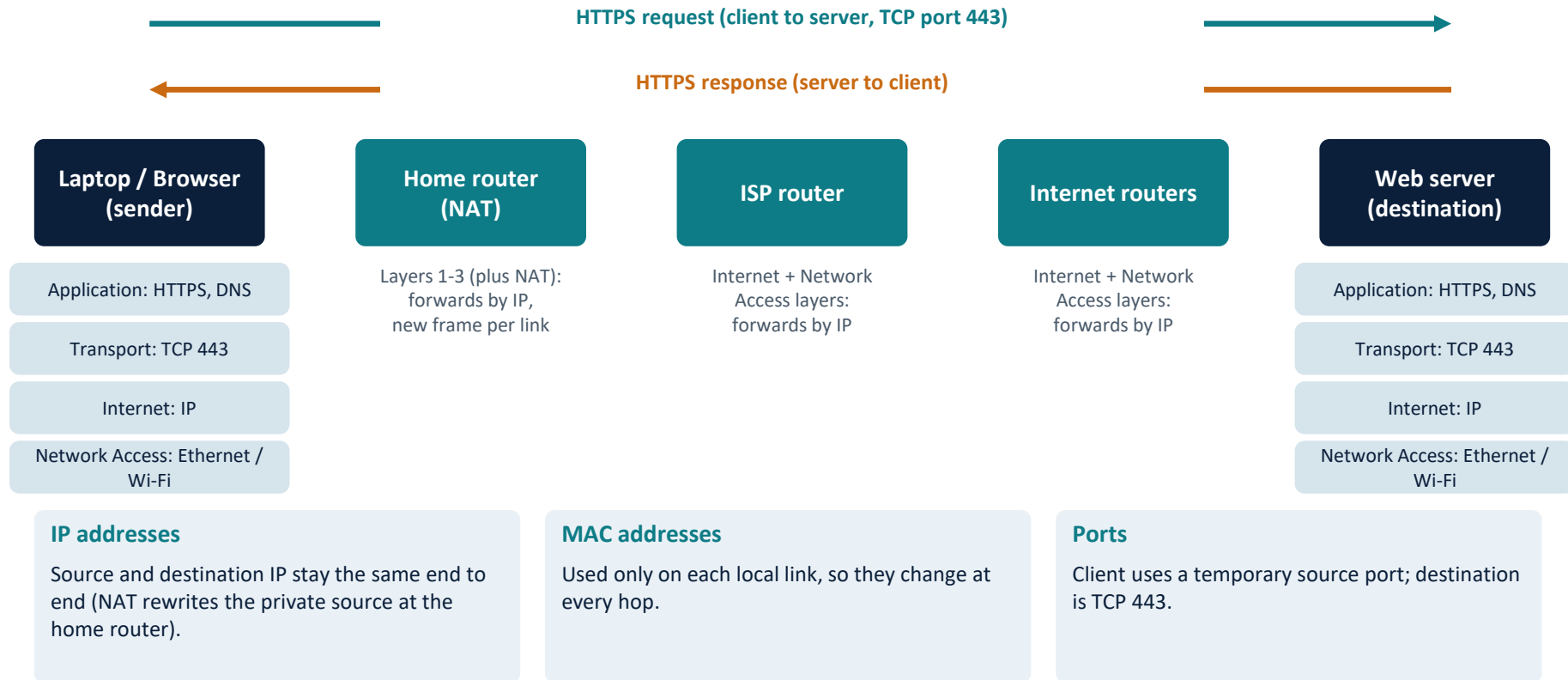
ENCAPSULATION



Sender: each layer adds a header as data goes down, then bits are sent. Receiver: decapsulation removes headers in reverse order, up to the application.

Original Packet Journey Diagram

PACKET JOURNEY



Protocol Analysis Table

PROTOCOLS

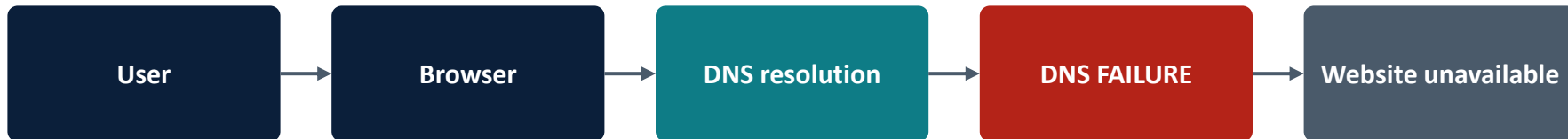
TCP/IP layer	Protocol	PDU	Addressing information	Purpose
Application	DNS	DNS message	Domain name	Resolves a domain name to an IP address
Application	HTTPS (HTTP over TLS)	HTTP message / TLS record	URL / domain name	Secure web request and response
Transport	TCP	Segment	Port numbers (443)	Reliable, ordered delivery
Transport	UDP	Datagram	Port numbers (53)	Low-overhead delivery (DNS queries)
Internet	IP	Packet	Source / destination IP	Routing between networks
Network Access	Ethernet / Wi-Fi	Frame	Source / destination MAC	Delivery on the local link
Network Access	ARP (IPv4)	ARP message	IP to MAC	Finds next-hop MAC address

Feature	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Acknowledgements, retransmission	No delivery guarantee
Ordering	Sequence numbers, ordered delivery	Not ordered
Overhead	Higher (20+ byte header, state)	Lower (8 byte header)
Examples	HTTPS (443), SSH (22)	DNS (53), online gaming, real-time media

UDP is not simply "faster": it has lower overhead and no TCP reliability mechanisms, so applications choose it for timeliness or build their own recovery. HTTP/3 uses QUIC over UDP.

Realistic Communication Failure: DNS Resolution Failure

FAILURE SCENARIO



- Symptom: `https://example.com` does not open, though other connectivity may work.
- Cause: the client cannot resolve the name, so no TCP connection is attempted.
- Realistic causes: unreachable DNS server, wrong client DNS setting, wrong DHCP or router DNS configuration.
- Key point: a DNS failure is not the same as a general internet connectivity failure.

1 Identify

ipconfig /all
Check IP, gateway and DNS servers.
Note symptom: site will not open.

2 Diagnose

ping 8.8.8.8 (IP path)
nslookup example.com (DNS)
IP works but DNS fails: check DNS server, client settings, router/DHCP.

3 Resolve

Correct the DNS configuration, or use a functioning resolver suited to the network.

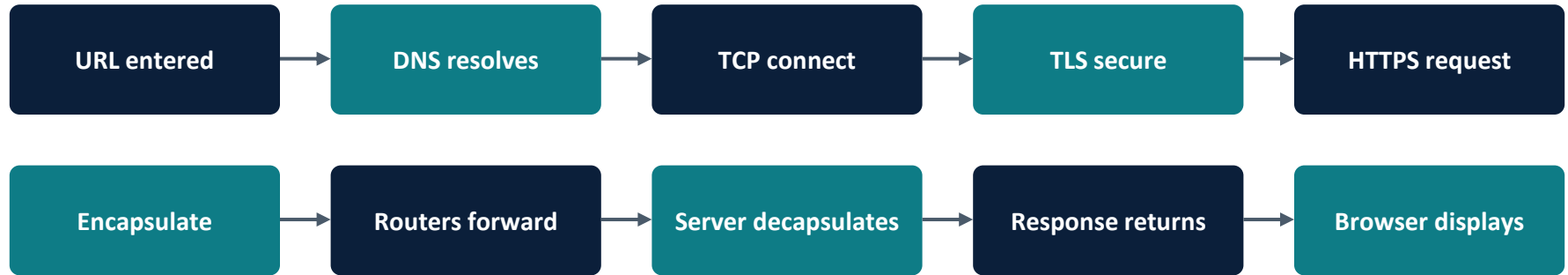
4 Verify

nslookup example.com
Reopen <https://example.com>

These are diagnostic tests: their results decide the diagnosis. A successful ping to 8.8.8.8 does not prove DNS works; nslookup testing the name does.

Complete Communication Process and Conclusion

SUMMARY



- Application: defines the HTTPS and DNS messages.
- Transport: delivers between processes using ports (TCP 443).
- Internet: IP addressing and routing between networks.
- Network Access: moves frames across each local link.

References

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AI-use declaration: An AI assistant was used to help organise the presentation, explain networking concepts and draft wording. I reviewed and verified the technical content against the references above and remain responsible for this submission.